

Analysis Amazon Data

Analysis Report

Prepared by Aseel

Executive Summary:

This report analyzes Amazon product data. The original file contains 1,465 rows and 13 columns; after cleaning, 1,398 products remain. The data covers 9 main categories and their subcategories.

The purpose is to two relationships: discount vs. rating, and discount vs. rating count— by main category and product. The analysis found an average discount of 47.27% and an average rating of 4.1. Note that some categories carry no discount, so the discount average does not cover all categories equally.

Scope:

This analysis is exploratory in nature. This analysis focuses on discovering two relationships by Main categories and product between rating vs discount percentage. The analysis is limited to categories, products, discount, rating count and rating. Analysis of review content and review title is outside the scope but will be addressed in a future analysis.

- Questions:
- How many products are there?
- How many Main categories are there?
- What is the average rating?
- What is the average discount percentage?
- What are the top 10 products by discount percentage?
- What are the top 10 products by rating count?
- What is the relationship between rating count and discount percentage per Main Category?
- What are the top 3 subcategories by average rating?
- What is the relationship between discount and rating per category?

Data Source:

The data was obtained from www.kaggle.com. The dataset includes the following columns: product_id, product_name, category, discounted_price, actual_price, discount_percentage, rating, rating_count (Number of reviews), user_id, user_name, review_id, review_title, and review_content.

I cleaned the data using Power Query. First, I removed duplicate rows. Then, I split the category column into six columns and renamed them to main category and subcategory 1 through subcategory 5. Next, I removed currency symbols from three quantitative columns. After that, I discovered an error in the discount_percentage column, so I recalculated it using a formula based on the discounted price and actual price.

The original data was kept separate from the cleaned version.

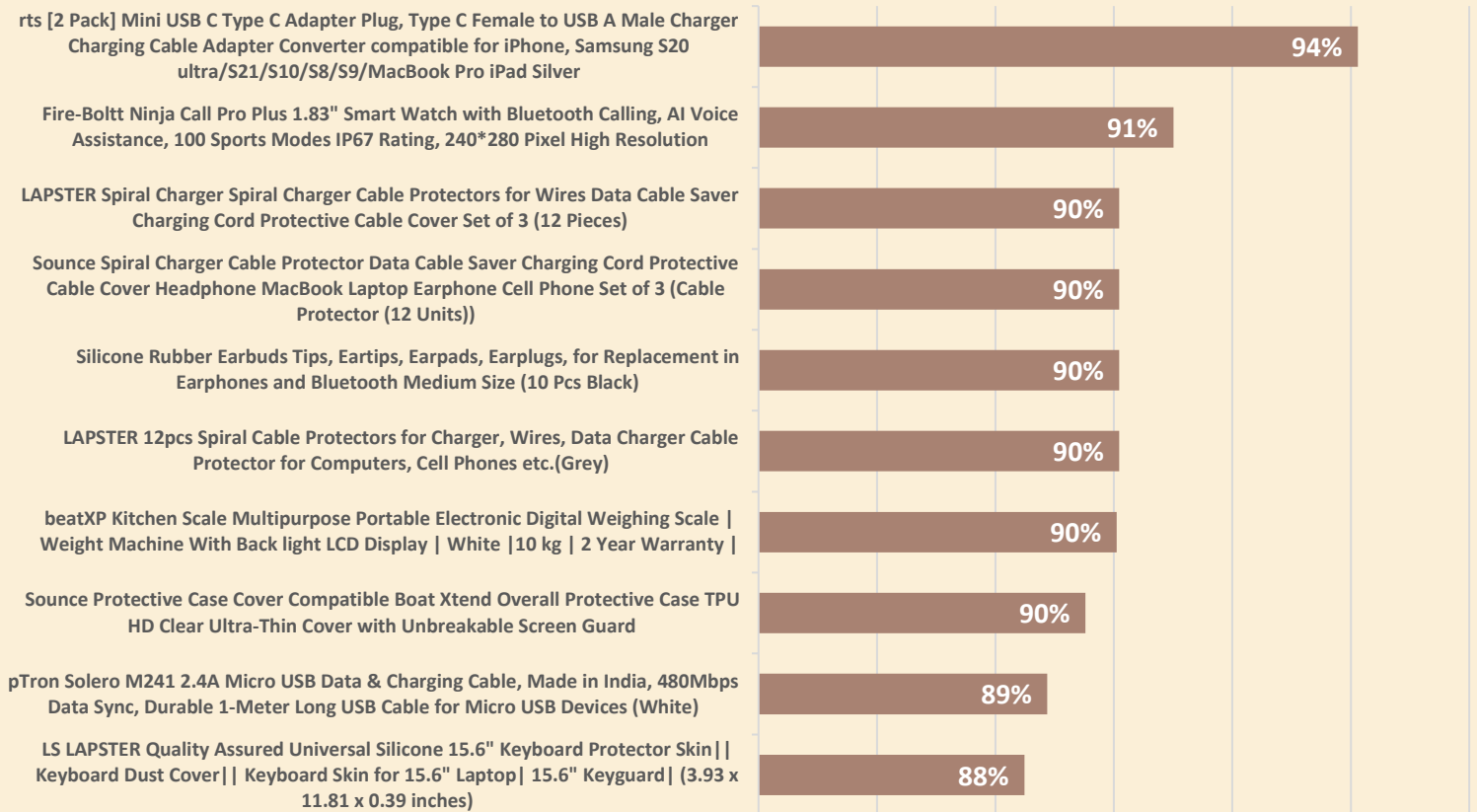
Analysis:

- How many products are there?
Number of products: 1398.
- How many Main categories are there?
Number of main categories: 9.
- What is the average rating?
Average rating: 4.1.
- What is the average discount percentage?
Average discount percentage: 47.27%.

- What are the top 10 products by discount percentage?

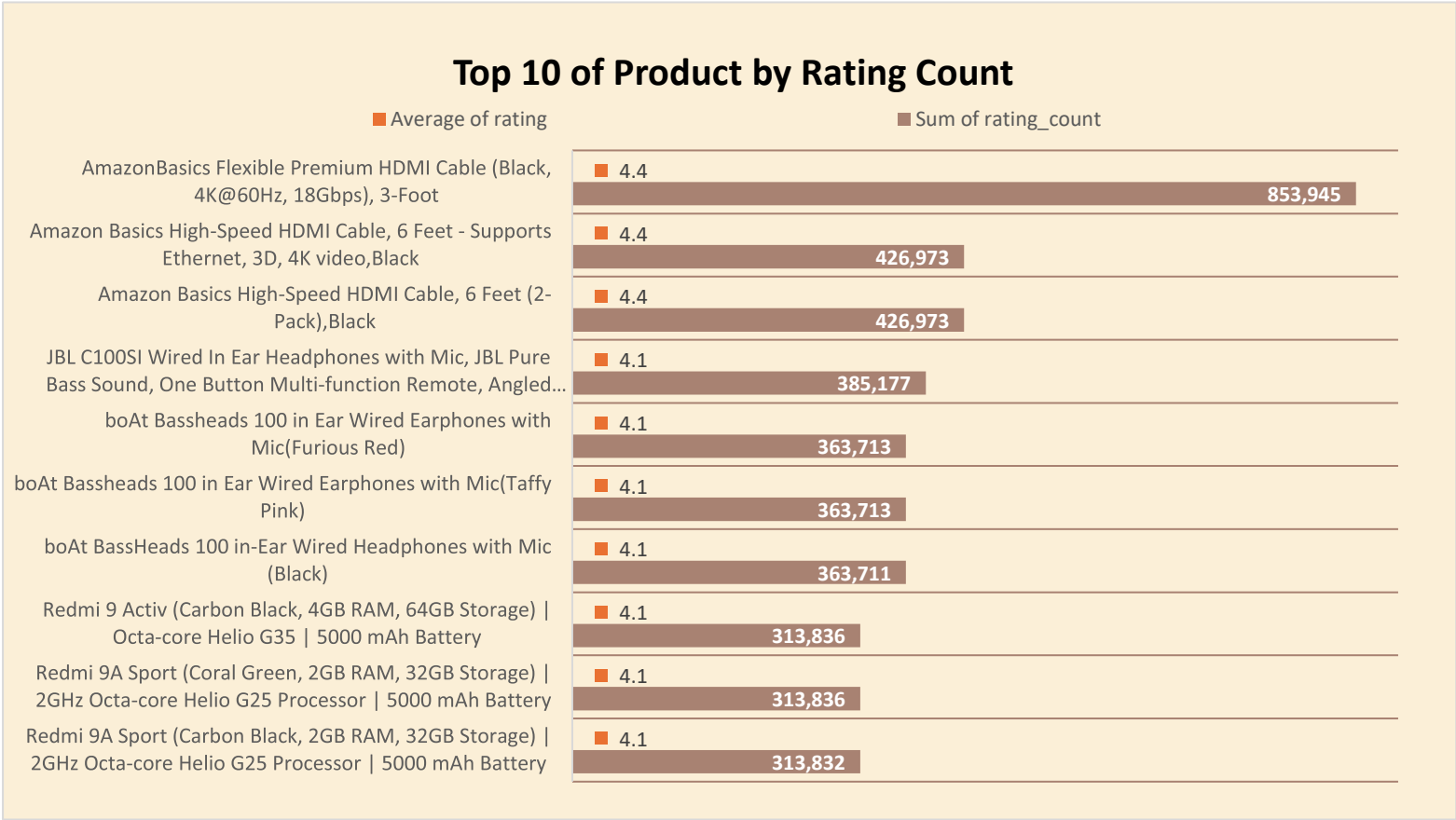
The product with the highest discount is (LS LAPSTER Quality Assured Universal Silicone 15.6" Keyboard Protector Skin|| Keyboard Dust Cover|| Keyboard Skin for 15.6" Laptop| 15.6" Keyguard| (3.93 x 11.81 x 0.39 inches))

Top 10 of Product By Average of Discount Percentage



- What are the top 10 products by rating count?

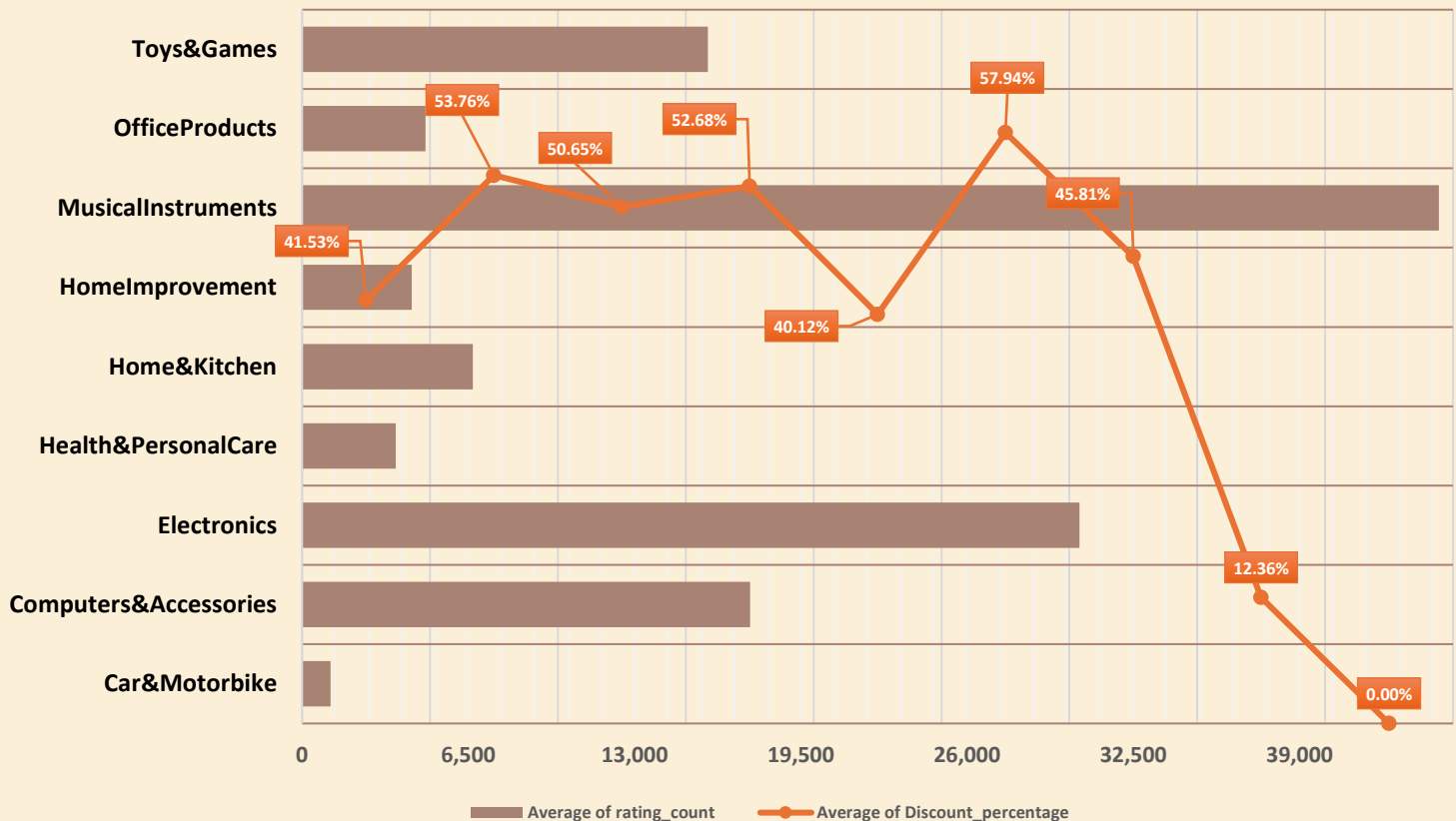
The product with the highest rating count is (Redmi 9A Sport (Carbon Black, 2GB RAM, 32GB Storage) | 2GHz Octa-core Helio G25 Processor | 5000 mAh Battery)



- What is the relationship between rating count and discount percentage per Main Category?

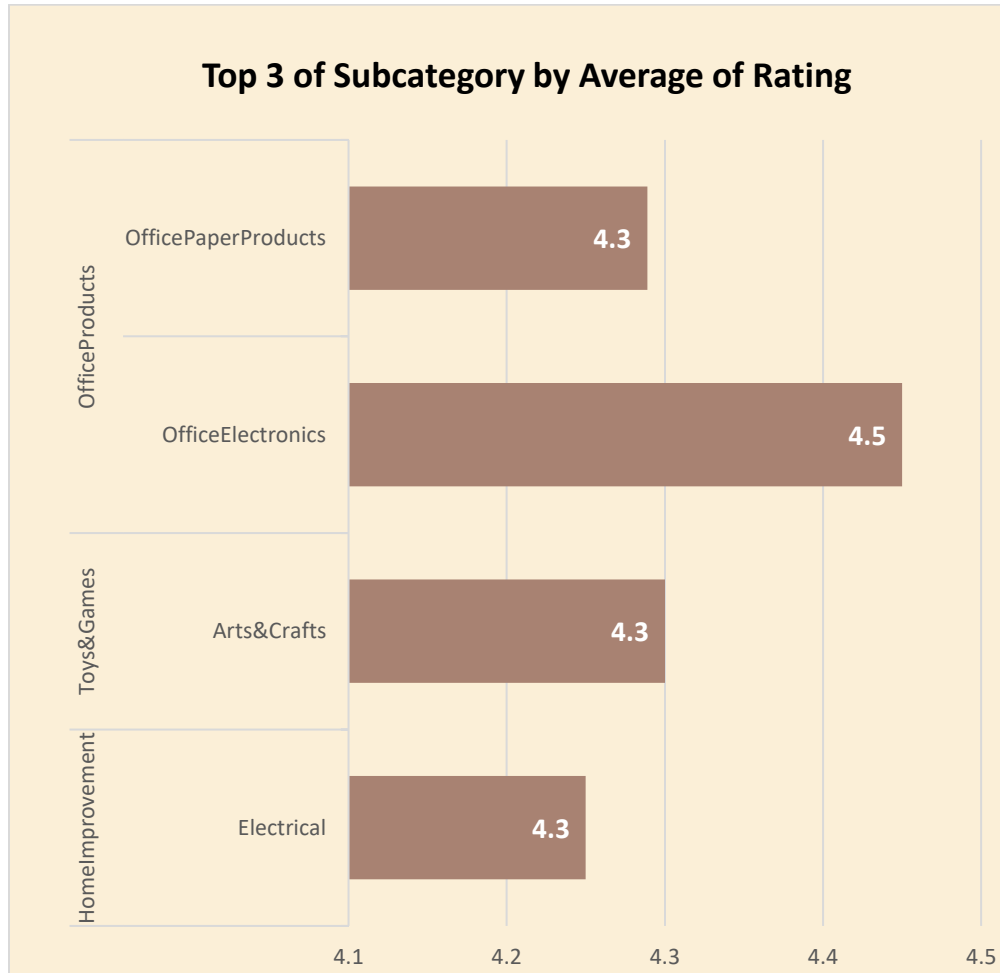
Actually, there is no clear relationship between rating count and discount percentage per main category, though discount has a slight effect.

Rating count VS Discount % for Main Category



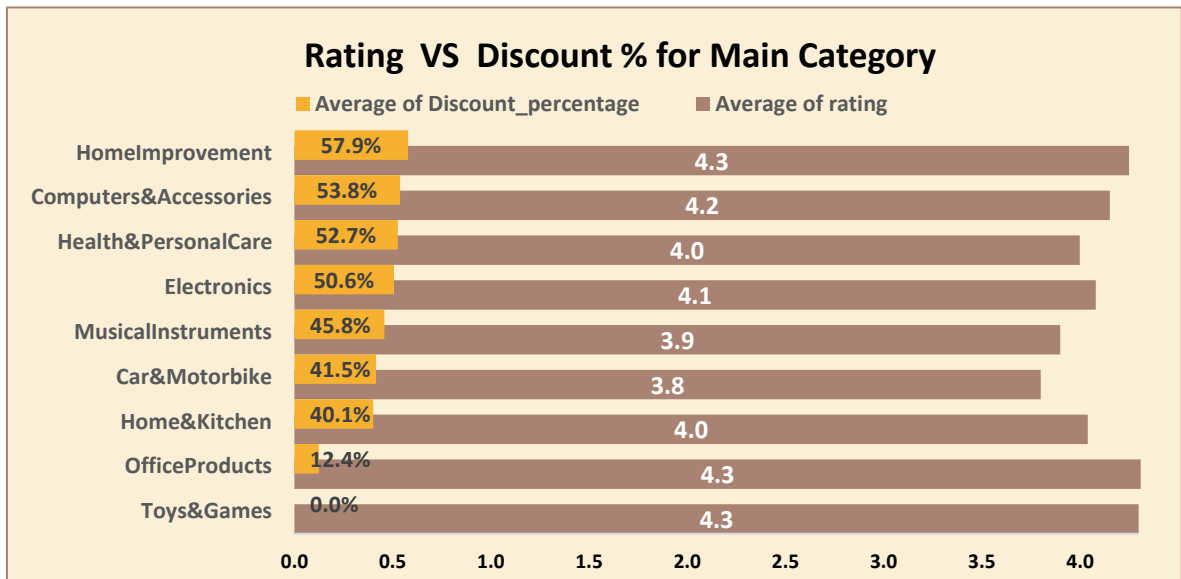
- What are the top 3 subcategories by average rating?

The subcategory with the highest average rating is Office Electronics from Office Products.



- What is the relationship between discount and rating per category?

The graph shows no clear relationship between rating and discount percentage for each main category, although the discount has a slight effect. This is due to the recurring rating of "4.3" and the 0.0% discount, as seen in the "Toys&Games" category.



Recommendations:

After analysis, it became clear that the discount percentage has minimal impact. The discount does not affect rating count or overall rating. For example, Toys&Games has a 0% discount yet a 4.3 rating — the same as categories with high discounts.. HomeImprovement, on the other hand, has the highest discount (58%), but its rating is also 4.3 — nearly identical despite the large difference in discount.

Therefore, the company should focus on product quality rather than relying on discounts to improve ratings. Further investigation of review_title and review_content is needed to identify what truly drives the rating.